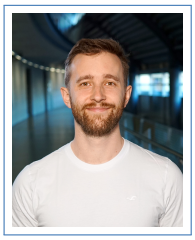
Lukas Burgholzer

Curriculum Vitae

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22 January 1994



Profile

Quantum-software researcher, engineering leader, and CTO working across the quantum computing software stack and HPCQC integration. I oversee the development of the *Munich Quantum Toolkit (MQT)* and the *Munich Quantum Software Stack (MQSS)*, including the Quantum Device Management Interface (QDMI), and co-founded MQSC. As of November 2026, my work has accumulated over **3,000 citations** (**h-index 31**); MQT has **4.6 million PyPI downloads** and **1.4k GitHub stars**.

Experience

2025–present **CTO and Founding Member of MQSC**, *Bootstrapped deep-tech startup that translates quantum-software research into industrial products*, `https://mq.sc`.

2024–present **Technical Lead of the Munich Quantum Software Stack (MQSS)**, *Munich Quantum Valley initiative with 400+ researchers; lead developer and maintainer of QDMI, an emerging de facto standard for interoperable quantum-hardware integration in HPCQC software stacks*, `https://github.com/Munich-Quantum-Software-Stack/QDMI`.

2023–present **Chief Developer of the Munich Quantum Toolkit (MQT)**, *Open-source ecosystem providing tools across the quantum computing software stack, including simulation, compilation, and verification; 4.6M PyPI downloads and 1.4k GitHub stars (November 2026)*, `https://mqt.readthedocs.io`.

2023–present **Senior Researcher**, *Technical University of Munich*, Germany.
Chair for Design Automation. Mentorship of 8+ doctoral researchers; initiated and co-developed courses, adopted by DTU and Northwestern University.

2019–2023 **Doctoral Researcher**, *Johannes Kepler University Linz*, Austria.
LIT Secure and Correct Systems Lab. Institute for Integrated Circuits.

2018–2019 **Graduate Research Associate**, *MathConsult GmbH*, Linz, Austria.

Education

2019–2023 **PhD Studies in Computer Science**, *Johannes Kepler University Linz*, Austria.
Supervisor: Univ.-Prof. Dr. Robert Wille (Institute for Integrated Circuits), *passed with distinction*

2017–2018 **Master's Degree in Industrial Mathematics**, *Johannes Kepler University Linz*, Austria.
passed with distinction

2016–2019 **Bachelor's Degree in Computer Science**, *Johannes Kepler University Linz*, Austria.
passed with distinction

2013–2016 **Bachelor's Degree in Mathematics**, *Johannes Kepler University Linz*, Austria.
passed with distinction

Skills

Languages	German (native), English (fluent)
Quantum software	Tools across the quantum computing software stack: simulation, compilation, verification, optimization, mapping, and hybrid execution
Algorithms	Decision diagrams, automated and optimized compilation, and scalable verification
Software engineering	Modern C++, Python, Git, CMake, and CI/CD
HPCQC and leadership	HPCQC integration and QDMI; technical leadership for MQT, MQSS, and MQSC; mentorship of doctoral researchers

Awards and Selected Achievements

2026 EuroTech Rising Stars Award, Rising Innovation Stars category, EuroTech Universities Alliance.

2026 High-Tech Young Talent Award, Bavarian State Ministry of Science and the Arts (Bavarian Hightech Awards), for research with particularly high innovation potential and strong prospects for practical application.

2025 EDAA Outstanding Dissertation Award, European Design and Automation Association.

2024 ACM SIGDA Outstanding PhD Dissertation Award.

2024 GI Dissertation Award nomination, Gesellschaft für Informatik, Austrian Computer Society, and Swiss Informatics Society.

2024 Heinz Zemanek Prize, Austrian Computer Society.

2024 PhD Forum Best Poster Prize, Design, Automation and Test in Europe (DATE).

2023 Best Paper Award nomination, Asia and South Pacific Design Automation Conference.

2022, 2023 Mentor, NYUAD International Hackathon for Social Good in the Arab World (quantum computing).

2022 JKU Young Researcher's Award, Johannes Kepler University Linz.

Research and Publications

Quantum computers are rapidly advancing, yet the corresponding software lags behind. Through my research, I bridge this gap by transferring methods from classical design automation to the quantum realm—creating the essential software layer that makes quantum hardware usable. My contributions include decision-diagram-based simulation methods that reduce workloads from days to minutes, automated and optimized compilation methods for diverse hardware platforms, compiler infrastructure based on MLIR, and efficient, scalable verification. I oversee the development of the *Munich Quantum Toolkit (MQT)* and the *Munich Quantum Software Stack (MQSS)*, including QDMI—an emerging de facto standard for interoperable quantum-hardware integration—ensuring that powerful quantum processors can be integrated into high-performance computing centers and utilized by end-users. My work covers the entire quantum computing software stack: from efficient data structures and simulation to compilation, verification, and hybrid execution.
I have published 80+ papers in leading venues including IEEE TCAD, ACM TQC, DAC, DATE, ICCAD, QCE, and SC. For a full list of publications, see `https://www.cda.cit.tum.de/team/burgholzer/`.

Open-Source Software

I am a strong advocate for open science. All methods developed in my research are released as part of the *Munich Quantum Toolkit (MQT)* (`https://github.com/munich-quantum-toolkit`), forming a comprehensive, open-source ecosystem for quantum software. These tools are engineered for performance (C++) and usability (Python), enabling thousands of researchers and developers worldwide to build quantum applications. The MQT has become a staple in the community, with **4.6 million downloads** on PyPI and **1.4k stars** on GitHub (as of November 2026). Tools natively integrate with IBM Qiskit, Xanadu's PennyLane, and `jeff`; they are deployed at HPC centers across Europe and beyond, including the Leibniz Supercomputing Centre (LRZ) and Oak Ridge National Laboratory, and serve as the technological backbone of MQSC.
For a full list of contributions, see `https://github.com/burgholzer`.